Applied Econometrics: Session 6

Matching and Propensity Scores

PhD Intensive Course

Lecture 6

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Why This Lecture Matters

Many empirical papers have a treatment group and a comparison group, but no random assignment.

Matching and propensity scores ask:

Core question

Can we make treated and untreated units comparable by conditioning carefully on pre-treatment observables?

- Matching tries to compare like with like.
- Propensity scores compress many observed covariates into one treatment probability.
- Neither method fixes selection on unobservables.

Roadmap for Today

1. Setup
2. Matching
3. Propensity Scores
4. Inverse Probability Weighting
5. Practical Workflow

The Observational Problem

We observe

$$(Y_i, D_i, X_i), \qquad D_i \in \{0, 1\}.$$

- Y_i: observed outcome.
- D_i: treatment indicator.
- X_i: pre-treatment covariates, such as age, education, past outcomes, location.

Potential outcomes:

$$Y_i = D_i Y_{1i} + (1 - D_i) Y_{0i}.$$

The problem is that treated and untreated people may differ in X_i before treatment.

Appendix details

The Required Assumption

Matching and propensity-score methods usually rely on conditional independence:

$$(Y_{1i}, Y_{0i}) \perp\!\!\!\perp D_i \mid X_i.$$

- After comparing units with the same X_i, treatment is as good as random.
- This is a strong economic assumption, not a theorem.
- X_i must be measured before treatment.
- Post-treatment variables are usually bad controls.

Plain English

Among people with the same observed background, the treated and untreated differ only because of treatment.

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Overlap: The Other Required Assumption

Conditional independence is not enough. We also need common support:

$$0 < \mathbb{P}(D_i = 1 \mid X_i = x) < 1 \qquad \text{for relevant } x.$$

- For each treated type, we need comparable untreated units.
- For each untreated type, we need comparable treated units if the target is the ATE.
- No estimator can learn a counterfactual from a covariate region containing only one treatment status.

Practical warning

Bad overlap is not a software problem. It changes what population can be studied credibly.

Appendix details

Which Effect Are We Estimating?

Three common targets:

$$\mathrm{ATE} = \mathbb{E}[Y_{1i} - Y_{0i}],$$

$$\mathrm{ATT} = \mathbb{E}[Y_{1i} - Y_{0i} \mid D_i = 1],$$

$$\mathrm{ATU} = \mathbb{E}[Y_{1i} - Y_{0i} \mid D_i = 0].$$

- Matching treated units to controls naturally estimates the ATT.
- Symmetric matching or IPW can target the ATE.
- With heterogeneous treatment effects, these are not the same object.

Matching: The Main Idea

- For each treated unit, find untreated units that look similar before treatment.
- Use those untreated outcomes as estimates of the treated unit's missing Y_{0i}.
- Then compare observed treated outcomes to imputed untreated outcomes.

Matching as design

We are trying to construct a comparison sample before looking too hard at outcomes.

Matching as estimation

We are imputing the missing potential outcome using observed units with similar X_i.

Exact Matching

Exact matching compares treated and untreated units in the same covariate cell.

$$\delta(x) = \mathbb{E}[Y_i \mid D_i = 1, X_i = x] - \mathbb{E}[Y_i \mid D_i = 0, X_i = x].$$

The ATT is a weighted average over treated cells:

$$\mathrm{ATT} = \sum_x \delta(x)\mathbb{P}(X_i = x \mid D_i = 1).$$

- Works well when X_i is low-dimensional and discrete.
- Fails quickly when X_i has many categories or continuous variables.

Appendix details

A Tiny Exact-Matching Example

Cell X	Treated	Control	Effect	Treated share	Contribution
Young, urban	12.0	9.5	2.5	0.40	1.00
Young, rural	10.0	8.0	2.0	0.20	0.40
Older, urban	14.0	12.5	1.5	0.25	0.38
Older, rural	11.0	10.0	1.0	0.15	0.15
			Exact-matching ATT		1.93

- We do not compare all treated to all controls.
- We compare within covariate cells and then average using the treated distribution.
- The weights matter because the target is the treated population.

Matching as Imputation

For a treated unit i, Y_{1i} is observed but Y_{0i} is missing. Matching estimates

$$\widehat{Y}_{0i} = \frac{1}{K} \sum_{\ell \in \mathcal{N}_K(i; D=0)} Y_\ell,$$

where $\mathcal{N}_K(i; D = 0)$ is the set of K nearest untreated neighbors. Then

$$\widehat{\mathrm{ATT}}_M = \frac{1}{N_1} \sum_{i:D_i=1} (Y_i - \widehat{Y}_{0i}).$$

- The missing counterfactual is filled in using nearby controls.
- Better neighbors reduce bias; more neighbors can reduce variance.

Appendix details

Nearest-Neighbor Matching

Exact matching is often impossible, so we use a distance.

$$d(i, \ell) = \left[(X_i - X_\ell)' W (X_i - X_\ell)\right]^{1/2}.$$

- $W = I$: Euclidean distance after standardizing covariates.
- $W = \widehat{\Sigma}_X^{-1}$: Mahalanobis distance.
- The closest untreated unit becomes the match for a treated unit.

Important detail

Always scale variables before distance matching. Otherwise income in euros can dominate age in years.

With Replacement or Without Replacement?

Choice	Advantage	Cost
With replacement	Best control can be reused, usually lowers bias	Fewer unique controls, often higher variance
Without replacement	Uses more control observations	Later treated units may receive poor matches
One nearest neighbor	Very local comparison	Noisy estimate
Many neighbors	Lower variance	More distant matches, more bias
Caliper	Blocks bad matches	Drops units outside tolerance

There is no universally correct setting. The choice should be reported and checked.

Bias-Variance Tradeoff in Matching

Matching is nonparametric. Its performance depends on how local the comparison is.

- Very strict matching: low bias, but many treated units may have no match and estimates are noisy.
- Loose matching: many matches and lower variance, but treated and control units may not be comparable.
- More covariates: controls more confounding, but makes good matches harder to find.

Practical rule

Matching quality is a design object. Inspect distances, covariate balance, and dropped observations.

Appendix details

The Curse of Dimensionality

Suppose each covariate is split into five bins.

$$\text{Number of cells} = 5^p,$$

where p is the number of covariates.

Covariates p	Cells 5^p	What happens
2	25	feasible
5	3,125	many empty cells
10	9,765,625	exact matching collapses

Lesson

The more dimensions we match on, the farther the nearest neighbor usually is.

Common Support in Matching

- If treated units are older, richer, and more educated than all controls, matching has no credible comparison.
- If some treated units have no nearby controls, the ATT for all treated units is not supported by the data.
- It may be better to report an effect for the matched or overlapping population.

Diagnostic questions

- Are matched control covariates close to treated covariates?
- How many treated units are unmatched or outside the caliper?
- Are results driven by a small set of reused controls?

What Matching Does Not Solve

Matching is not magic.

- It does not remove bias from unobserved ability, motivation, risk, or expectations.
- It does not make post-treatment controls safe.
- It does not prove that all relevant confounders were measured.
- It can increase bias if the distance metric prioritizes irrelevant variables.

Interpretation

A matching estimate is causal only if the remaining assignment variation within matched sets is plausibly as good as random.

Why Propensity Scores?

Matching on many covariates is hard. Propensity scores compress treatment selection into one number:

$$e(X_i) = \mathbb{P}(D_i = 1 \mid X_i).$$

- $e(X_i)$ is the probability of being treated for a unit with covariates X_i.
- Units with the same propensity score have the same predicted treatment probability.
- We can match, stratify, trim, or weight using $e(X_i)$.

Intuition

Instead of matching on the full profile, match on how likely the unit was to be treated.

The Propensity-Score Theorem

If treatment is conditionally independent of potential outcomes given X_i,

$$(Y_{1i}, Y_{0i}) \perp\!\!\!\perp D_i \mid X_i,$$

then treatment is also conditionally independent given the propensity score:

$$(Y_{1i}, Y_{0i}) \perp\!\!\!\perp D_i \mid e(X_i).$$

- The propensity score is a balancing score.
- We can condition on a scalar instead of the whole vector X_i.
- This does not mean the propensity-score model can be sloppy.

Appendix details

Balancing Score Intuition

If two units have the same $e(X)$, then their covariates may differ, but their predicted treatment chance is the same.

Within a narrow propensity-score bin:

$$\mathbb{P}(D_i = 1 \mid e(X_i) = p) = p.$$

- Treatment should not be systematically related to the covariates used to estimate $e(X)$.
- We check this by comparing covariate balance within score strata or after weighting.
- If balance is poor, revise the score model or change the design.

Beginner trap

A high predictive AUC is not the goal. Balance and overlap are the goals.

Estimating the Propensity Score

Common first step:

$$\widehat{e}(X_i) = \Lambda(X_i'\widehat{\gamma}), \qquad \Lambda(z) = \frac{\exp(z)}{1+\exp(z)}.$$

- Use only pre-treatment variables.
- Include variables that affect both treatment and outcome.
- Flexible terms can help balance: interactions, squares, splines, or machine learning.
- Do not include instruments merely because they predict treatment; they can worsen overlap.

The score model is a design model

Its purpose is to create comparable groups, not to explain treatment choice beautifully.

Overlap Diagnostics

After estimating $\widehat{e}(X)$, inspect:

- Histograms or density plots by treatment status.
- Minimum and maximum score in treated and untreated groups.
- Fraction of observations near 0 or 1.
- Effective sample size after weighting.
- Standardized mean differences before and after adjustment.

Standardized mean difference

$$\mathrm{SMD}(X_j) = \frac{\bar{X}_{j,1} - \bar{X}_{j,0}}{s_j}.$$

Values close to zero indicate better balance.

Trimming

Trimming removes observations with weak overlap.

$$a \leq \widehat{e}(X_i) \leq b.$$

Common examples:

$$0.05 \leq \widehat{e}(X_i) \leq 0.95 \qquad \text{or} \qquad \max\{\min \widehat{e}_1, \min \widehat{e}_0\} \leq \widehat{e}_i \leq \min\{\max \widehat{e}_1, \max \widehat{e}_0\}.$$

- Trimming can reduce extrapolation and extreme weights.
- It changes the target population.
- Always report how many observations were dropped.

Appendix details

Propensity-Score Matching

Use the scalar score instead of the full covariate vector.

$$d(i, \ell) = |\widehat{e}(X_i) - \widehat{e}(X_\ell)|.$$

- Treated units are matched to controls with similar estimated treatment probability.
- Calipers are easy: only accept matches within a score distance.
- Balance should still be checked on the original covariates, not only on the score.

Key limitation

Two units with the same score can have different covariate profiles. The theorem is about distributions, not about perfect twins.

IPW: Reweighting the Sample

Inverse probability weighting gives more weight to observations that are rare for their covariates.

$$w_i^{ATE} = \frac{D_i}{e(X_i)} + \frac{1 - D_i}{1 - e(X_i)}.$$

- Treated unit with low $e(X)$: rare treated type, large weight.
- Control unit with high $e(X)$: rare control type, large weight.
- Extreme scores create extreme weights.

Intuition

Weighting reconstructs a pseudo-population where treatment no longer depends on observed covariates.

Appendix details

IPW Estimator for the ATE

A normalized, or Hajek-style, version is

$$\widehat{\mathrm{ATE}}_{IPW} = \frac{\sum_i D_i Y_i / \widehat{e}(X_i)}{\sum_i D_i / \widehat{e}(X_i)} - \frac{\sum_i (1 - D_i) Y_i / [1 - \widehat{e}(X_i)]}{\sum_i (1 - D_i) / [1 - \widehat{e}(X_i)]}.$$

- The treated group is weighted to represent everyone.
- The untreated group is weighted to represent everyone.
- The difference estimates the population average treatment effect under CIA and overlap.

IPW Estimator for the ATT

For the treated population, treated units already represent treated units. Controls must be reweighted to look like the treated:

$$w_i^{ATT} = \begin{cases} 1, & D_i = 1, \\ \widehat{e}(X_i)/[1 - \widehat{e}(X_i)], & D_i = 0. \end{cases}$$

Then

$$\widehat{\mathrm{ATT}}_{IPW} = \frac{\sum_i D_i Y_i}{\sum_i D_i} - \frac{\sum_i (1 - D_i) w_i^{ATT} Y_i}{\sum_i (1 - D_i) w_i^{ATT}}.$$

- Controls who looked likely to be treated receive larger weight.
- This targets the treated group, not the full population.

Weights Need Diagnostics

- Report the largest weights and percentiles of weights.
- Check effective sample size:

$$N_{\text{eff}} = \frac{(\sum_i w_i)^2}{\sum_i w_i^2}.$$

- Compare covariate balance before and after weighting.
- Consider trimming or overlap weights when extreme weights dominate.

Interpretation problem

If 2,000 observations behave like 60 effective observations after weighting, the design is fragile even if the point estimate looks precise.

A Practical Workflow

1. Define the target estimand: ATE, ATT, or an overlap population effect.
2. Choose pre-treatment covariates using economic reasoning.
3. Inspect raw treatment-control differences in covariates.
4. Estimate a matching or propensity-score design.
5. Check balance, overlap, dropped units, and weights.
6. Estimate treatment effects.
7. Report uncertainty and sensitivity to design choices.

Order matters

Diagnostics should come before celebrating the treatment-effect estimate.

Python Example for This Lecture

The companion script is:

`lectures/code/lecture-6-matching-propensity.py`.

It simulates a setting where:

- treatment is more likely for some observed covariate profiles;
- treatment effects are heterogeneous;
- the true ATE and ATT are known from simulated potential outcomes;
- naive, regression-adjusted, nearest-neighbor matched, and propensity-score weighted estimates are compared.

Purpose

The script is not a recipe for publication. It is a controlled lab for seeing selection bias, matching, overlap, and weighting.

What the Script Demonstrates

Estimate	What students should look for
Naive treated-control gap	Combines treatment effect and selection bias.
Regression adjusted	Works well when the conditional mean is modeled adequately.
Nearest-neighbor matching	Targets the ATT by imputing missing untreated outcomes for treated units.
Propensity-score IPW	Reweights observations to target the ATE or ATT.
Trimmed IPW	Shows how overlap restrictions can stabilize weights and change the sample.

The same data can give different estimates because methods target different populations and use different modeling assumptions.

How to Present a Matching or Propensity-Score Result

Minimum reporting:

- Causal estimand and identifying assumption.
- Covariates used and why they are pre-treatment.
- Matching method or propensity-score model.
- Balance before and after adjustment.
- Overlap plots or score ranges.
- Number of observations dropped or reused.
- Main estimate, standard error, and sensitivity checks.

Reader question

Would I believe treated and comparison units are exchangeable after this adjustment?

Common Failure Modes

- Matching on variables affected by treatment.
- Treating a propensity-score regression table as evidence of causal validity.
- Reporting no balance table or overlap diagnostic.
- Estimating an ATE when there is overlap only for the treated population.
- Keeping extreme weights because the point estimate is convenient.
- Using matching to hide a weak research design.

Bottom line

Matching and propensity scores can make selection-on-observables designs more transparent, but they do not create unobserved counterfactuals.

Takeaways

1. Matching compares treated and untreated units with similar pre-treatment covariates.
2. Exact matching is clear but usually infeasible in high dimensions.
3. Nearest-neighbor matching is imputation of missing potential outcomes.
4. Propensity scores reduce covariates to a scalar balancing score.
5. Overlap is empirical and must be diagnosed.
6. IPW creates a pseudo-population but can be unstable with extreme scores.
7. The causal assumption remains selection on observables.

Sources and Further Reading

- Matheus Facure Alves, *Causal Inference for the Brave and True*, Chapter 10, Matching: `https://matheusfacure.github.io/python-causality-handbook/10-Matching.html`
- Matheus Facure Alves, *Causal Inference for the Brave and True*, Chapter 11, Propensity Score: `https://matheusfacure.github.io/python-causality-handbook/11-Propensity-Score.html`
- Rosenbaum, P. R. and Rubin, D. B. (1983). "The central role of the propensity score in observational studies for causal effects."
- Angrist, J. D. and Pischke, J.-S. (2009). *Mostly Harmless Econometrics*.

Appendix Map

- A1: Conditional independence and matching identification.
- A2: Exact matching as the g-formula.
- A3: Nearest-neighbor matching as imputation.
- A4: Bias from imperfect matches.
- A5: Propensity-score theorem.
- A6: IPW identities.
- A7: Overlap, trimming, and target populations.

Appendix A1: CIA and Matching Identification

Under conditional independence,

$$(Y_{1i}, Y_{0i}) \perp\!\!\!\perp D_i \mid X_i.$$

Therefore, for $d \in \{0, 1\}$,

$$\mathbb{E}[Y_{di} \mid X_i = x, D_i = 1] = \mathbb{E}[Y_{di} \mid X_i = x, D_i = 0] = \mathbb{E}[Y_{di} \mid X_i = x].$$

Since observed outcomes reveal the realized potential outcome,

$$\mathbb{E}[Y_i \mid X_i = x, D_i = d] = \mathbb{E}[Y_{di} \mid X_i = x, D_i = d] = \mathbb{E}[Y_{di} \mid X_i = x].$$

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Appendix A2: Exact Matching as the G-Formula

For the ATT,

$$\mathrm{ATT} = \mathbb{E}[Y_{1i} - Y_{0i} \mid D_i = 1].$$

Apply iterated expectations over X_i:

$$\mathrm{ATT} = \sum_x \{\mathbb{E}[Y_{1i} \mid X_i = x, D_i = 1] - \mathbb{E}[Y_{0i} \mid X_i = x, D_i = 1]\}\, \mathbb{P}(X_i = x \mid D_i = 1).$$

By CIA and consistency,

$$\mathrm{ATT} = \sum_x \{\mathbb{E}[Y_i \mid X_i = x, D_i = 1] - \mathbb{E}[Y_i \mid X_i = x, D_i = 0]\}\, \mathbb{P}(X_i = x \mid D_i = 1).$$

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Appendix A3: Nearest-Neighbor Matching as Imputation

For each treated unit i, define weights over controls:

$$W_{i\ell} = \begin{cases} 1/K, & \ell \in \mathcal{N}_K(i; D = 0), \\ 0, & \text{otherwise.} \end{cases}$$

The imputed untreated outcome is

$$\widehat{Y}_{0i} = \sum_{\ell : D_\ell = 0} W_{i\ell} Y_\ell.$$

The ATT estimator is

$$\widehat{\mathrm{ATT}}_M = \frac{1}{N_1} \sum_{i : D_i = 1} \left(Y_i - \sum_{\ell : D_\ell = 0} W_{i\ell} Y_\ell \right).$$

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Appendix A4: Bias from Imperfect Matches

Let $m_0(x) = \mathbb{E}[Y_{0i} \mid X_i = x]$. For a treated unit i, its matched control has covariates $X_{m(i)}$. The expected imputation error is approximately

$$\mathbb{E}[Y_{0,m(i)} - Y_{0i} \mid X_i, X_{m(i)}] = m_0(X_{m(i)}) - m_0(X_i).$$

If $m_0(\cdot)$ is smooth,

$$m_0(X_{m(i)}) - m_0(X_i) \approx \nabla m_0(X_i)'(X_{m(i)} - X_i).$$

- Bias shrinks when matches are close.
- In high dimensions, nearest neighbors are often not close.

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Appendix A5: Proof of the Propensity-Score Theorem

Let $e(X_i) = \mathbb{P}(D_i = 1 \mid X_i)$. First, $D_i \perp\!\!\!\perp X_i \mid e(X_i)$ in the balancing-score sense because

$$\mathbb{P}(D_i = 1 \mid X_i, e(X_i)) = e(X_i)$$

and

$$\mathbb{P}(D_i = 1 \mid e(X_i)) = e(X_i).$$

If $(Y_{1i}, Y_{0i}) \perp\!\!\!\perp D_i \mid X_i$, then for $d \in \{0, 1\}$,

$$\mathbb{P}(D_i = 1 \mid Y_{di}, X_i) = \mathbb{P}(D_i = 1 \mid X_i) = e(X_i).$$

Averaging over all X_i values with the same score gives

$$\mathbb{P}(D_i = 1 \mid Y_{di}, e(X_i)) = e(X_i) = \mathbb{P}(D_i = 1 \mid e(X_i)).$$

Hence $Y_{di} \perp\!\!\!\perp D_i \mid e(X_i)$.

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Appendix A6: IPW Identities

Under CIA and overlap,

$$\mathbb{E}\left[\frac{D_i Y_i}{e(X_i)}\right] = \mathbb{E}\left[\frac{D_i Y_{1i}}{e(X_i)}\right].$$

Condition on X_i:

$$\mathbb{E}\left[\frac{D_i Y_{1i}}{e(X_i)} \mid X_i\right] = \frac{\mathbb{E}[D_i \mid X_i]\mathbb{E}[Y_{1i} \mid X_i]}{e(X_i)} = \mathbb{E}[Y_{1i} \mid X_i].$$

Therefore

$$\mathbb{E}\left[\frac{D_i Y_i}{e(X_i)}\right] = \mathbb{E}[Y_{1i}].$$

Similarly,

$$\mathbb{E}\left[\frac{(1-D_i) Y_i}{1-e(X_i)}\right] = \mathbb{E}[Y_{0i}].$$

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Appendix A7: Overlap, Trimming, and Target Populations

Suppose we keep only observations with $a \leq e(X_i) \leq b$. The estimand becomes

$$\mathbb{E}[Y_{1i} - Y_{0i} \mid a \leq e(X_i) \leq b].$$

This may be more credible than the full ATE when overlap is weak.

For a weighted estimator, extreme scores inflate variance:

$$\mathrm{Var}\left(\frac{D_i Y_i}{e(X_i)}\right)$$

becomes large when $e(X_i)$ is close to zero.

Conclusion

Trimming is not just a numerical fix. It defines the population for which the data contain credible treated-control comparisons.

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